PRODUCT REQUIREMENTS DOCUMENT • VERSION 1.0

MILO

A small island for your Windows desktop.
A calmer, more delightful way to reach the things you need.

Full Project Product Requirements Document

Windows-first • Downloadable desktop application • All-in-one desktop companion

Prepared for product design, engineering, QA, release, and ongoing product decisions

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Document control

Field	Value
Product	MILO
Document	Full Product Requirements Document (PRD)
Version / status	1.0 / Baseline specification for implementation
Primary platform	Windows desktop (Windows 10 1809+ technical floor for media-session capability; validate final supported OS matrix before release)
Product form	Installed, downloadable desktop application; not a website or browser-only experience
Primary audience	Individual Windows users seeking a lightweight, ambient desktop command surface
Related artifact	MILO — Full Technical Requirements Document v2.0
Document owner	Product owner / project lead
Review cadence	At each milestone and before scope-changing decisions

How to use this document

This PRD defines what MILO is, who it serves, the user problems it solves, how the product should behave, and how success will be evaluated. It is the product-level source of truth. The technical requirements document defines implementation architecture, native integration details, interfaces, data structures, and engineering constraints. Where a product requirement depends on an OS capability, engineering must validate the capability on the supported Windows versions and hardware matrix.

Normative language: MUST indicates a release-blocking requirement; SHOULD indicates a strong expectation with an approved exception path; MAY indicates an optional capability. Requirements are uniquely identified for traceability.

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1. Executive summary

MILO is a Windows-first desktop companion built around a distinctive floating island attached to the top-center edge of the user's screen. In its resting state, MILO is compact and unobtrusive. Hovering over or deliberately activating it reveals an expanded deck that grows down into the desktop and presents a carefully organised set of everyday controls and lightweight tools.

The product combines three roles that are usually scattered across different places: a glanceable system-control surface, a personal productivity space, and a temporary file handoff area. It also includes a music surface with lyrics when supported and a small animated companion that gives the experience warmth without becoming noisy or demanding.

MILO is not intended to replace Windows Settings, File Explorer, a full task manager, a full calendar, or a music streaming service. It is a fast-access layer that makes a curated set of frequent actions and information feel close at hand.

Product promise

MILO should feel like a little place on the desktop that is always nearby, never in the way, and genuinely useful within seconds.

Core experience

- Collapsed: a slim, elegant island at the top-center screen edge, visible enough to discover and small enough to ignore.
- Expanded: a smooth, layered deck with a concise overview and scrollable sections for system controls, productivity, files, and music.
- Responsive: adapts to display bounds, scaling, taskbar placement, fullscreen contexts, and multi-monitor setups.
- Personal: a cute avatar appears only in the expanded state and responds subtly to meaningful interactions.
- Trustworthy: local-first by default, transparent about permissions and limitations, and never silently moves or uploads user files.

Primary value

Reduce the effort of context switching. Users should be able to check the time, see their battery state, pause music, adjust supported controls, add a reminder, start a timer, or temporarily hold a file without opening a large application or losing their current workflow.

Release definition

The first public release is a polished Windows desktop application with a stable island shell, reliable system status, basic controls, reminders/timers/to-dos, drag-out file holding, current media controls and metadata, a restrained avatar system, robust settings, and an installer/update path. Advanced integrations that depend on hardware, driver, provider, or OS support must degrade gracefully rather than block the core experience.

2. Product vision and principles

2.1 Vision

Create an ambient desktop companion that makes useful actions feel effortless. MILO should be visually memorable, but its value must come from utility, speed, and dependable behaviour—not decoration alone.

2.2 Product principles

Principle	What it means in practice
Useful before cute	The companion and visual effects support the workflow; they never obstruct controls or consume attention unnecessarily.
Reveal, don't occupy	The default state is compact. Rich controls appear on intent and retreat when the user is done.
Fast paths first	Common tasks require few interactions. The overview shows the most relevant information without forcing navigation.
Local-first trust	Personal reminders, lists, and held-file metadata remain on-device unless a clearly disclosed feature requires otherwise.
Honest capability	If Windows, hardware, drivers, or an external player do not expose a capability, MILO explains that limitation and offers a sensible fallback.
Quiet by default	No unnecessary sounds, pop-ups, promotional prompts, or persistent avatar interruptions.
Consistent interaction	Hover, click, keyboard, touchpad, drag, focus, and escape behaviours follow predictable rules.
Graceful resilience	MILO remains useful when media, network, Bluetooth, lyrics, or a particular hardware control is unavailable.

2.3 Experience qualities

- Elegant and futuristic, but warm rather than cold.
- Dark, luminous visual identity with rich surfaces, restrained glow, and gradient accents.
- Soft organic curves and layered glass-like depth, balanced with legible contrast.
- Motion that communicates state and spatial relationships rather than adding spectacle.
- Low cognitive load: clear labels, familiar icons, predictable grouping, and progressive disclosure.

2.4 Product north star

A user can complete a frequent desktop micro-task from MILO faster and with less disruption than through their current workflow, while still feeling that the desktop remains theirs.

3. Problem statement and opportunity

3.1 User problems

- Everyday system information is fragmented across the taskbar, quick settings, Settings, and individual applications.
- Small productivity actions—setting a timer, capturing a reminder, checking a to-do—often require opening a separate app and breaking focus.
- Moving a file between applications can involve repeated File Explorer navigation, file pickers, and window switching.
- Media control and now-playing context are separated from the user's active work, and lyrics may be unavailable or buried in the player.
- Desktop utility apps often feel either visually generic or overly intrusive; users want an attractive, useful layer that does not take over the screen.

3.2 Opportunity

MILO can unify a narrow, high-frequency set of desktop actions in a single spatially memorable surface. Its differentiator is not feature breadth alone; it is the combination of top-edge presence, progressive expansion, integrated micro-tools, temporary file holding, and a friendly companion.

3.3 Product hypothesis

If MILO offers the most commonly needed desktop micro-actions in a compact, instantly accessible island, then users will rely on it as a habitual desktop utility, increasing task completion speed and perceived desktop convenience without increasing distraction.

3.4 Assumptions to validate

- Users understand the top-center island affordance without lengthy onboarding.
- Hover-to-expand is discoverable but does not trigger accidentally during normal pointer movement.
- Users value the combination of system controls, productivity, and file holding enough to keep MILO installed.
- An avatar can add emotional character when restrained and optional.
- Windows APIs and hardware support are sufficiently consistent for the target device matrix, or fallbacks can preserve the core value.

4. Goals, non-goals, and product boundaries

4.1 Goals

- G1 — Provide a compact, always-nearby desktop island with reliable expand/collapse behaviour.
- G2 — Provide a glanceable system deck for time/date, battery/power, media state, and supported quick controls.
- G3 — Provide lightweight productivity tools: reminders, timers, alarms, and to-do lists.
- G4 — Provide a temporary dropzone that can hold files and support deliberate drag-out to other applications.
- G5 — Provide current media metadata and transport controls, with lyrics when a supported source is available.
- G6 — Deliver a distinctive visual identity and a companion avatar that is expressive but unobtrusive.
- G7 — Meet baseline accessibility, privacy, security, and performance expectations for a background desktop utility.
- G8 — Ship as a signed, installable Windows desktop application with clear update and uninstall paths.

4.2 Non-goals for initial release

- Replacing Windows Settings, Action Center, File Explorer, Task Manager, or the system clock.
- Full calendar, email, notes, project management, or cloud task synchronisation.
- Becoming a music player or streaming service; MILO controls supported external sessions.
- Downloading, storing, or redistributing copyrighted lyrics without appropriate rights and provider terms.
- Acting as a general file manager, permanent cloud drive, or automatic upload agent.
- System-wide automation, arbitrary scripting, registry cleaning, driver management, or overclocking.
- Mobile, macOS, or Linux support in the first release.
- Always-on voice assistant, microphone capture, camera access, or behavioural surveillance.

4.3 Product boundary

MILO is an overlay-like desktop utility with a local application data store and narrowly scoped native integrations. It should not intercept unrelated user input, change system settings without explicit action, or claim to control a system feature that is unavailable on the device.

4.4 Definition of the all-in-one scope

The all-in-one scope means that the named MILO modules ship within one installed product and one coherent interface. It does not imply that every Windows hardware function is universally controllable, nor that every media provider exposes the same metadata or lyric rights.

5. Users, personas, and jobs to be done

5.1 Primary audience

Windows users who spend substantial time at a desktop or laptop and frequently move between work, browsing, communication, and media. They appreciate thoughtful interface design and want useful tools to be close without keeping multiple windows open.

5.2 Personas

Persona A — The focused worker

Works in a browser, IDE, design tool, or office suite. Needs timers, reminders, task capture, current time, and media controls. Values speed and low distraction.

Persona B — The multitasking student

Moves between documents, learning platforms, notes, and music. Needs quick timers, alarms, short task lists, and a way to park files while switching contexts.

Persona C — The desktop power user

Uses keyboard shortcuts, multiple displays, and many applications. Wants predictable behaviour, configurable hotkeys, no focus theft, and controls that do not fight the operating system.

Persona D — The aesthetic customiser

Cares about visual polish, themes, animations, and small details. Wants MILO to feel like part of the desktop environment while retaining control over intensity and visibility.

5.3 Jobs to be done

- When I am focused on another app, I want to glance at time and battery so I can stay oriented without switching windows.
- When music is playing, I want to pause or skip and see what is playing without opening the player.
- When I remember a task, I want to capture it quickly so I do not lose the thought.
- When I need to wait for something, I want to start a timer in seconds and trust that it will alert me.
- When I am uploading or moving between apps, I want to temporarily hold files so I can retrieve them from one convenient place.
- When I want to adjust a supported system control, I want a clear, direct control with feedback about whether the change succeeded.

5.4 Accessibility of personas

Personas describe contexts, not ability assumptions. MILO must work for keyboard-only users, users with low vision, users using screen readers, users sensitive to motion, and users who cannot rely on hover.

6. Product positioning and experience model

6.1 Positioning

MILO is a desktop companion and quick-access island—not a launcher, not a dock replacement, and not a full dashboard that demands constant attention.

6.2 Mental model

The island is a small entry point into a larger deck. The collapsed island signals availability. Expansion reveals the deck. The deck contains sections, each with a clear purpose. The user can glance, act, and leave.

6.3 Experience states

State	Description	Expected behaviour
Collapsed	Compact top-center island	Low visual footprint; no focus capture; hover/keyboard activation available.
Expanding	Transition into deck	Smooth spatial expansion; pointer remains tracked; no content jumps.
Expanded	Deck visible and interactive	Sections and controls accessible; avatar visible; focus behaviour predictable.
Pinned	User intentionally keeps deck open	Deck remains open until dismissed or pin is toggled off.
Compact focus	User interacts with a control that should not require full deck	Relevant subpanel may remain open while unrelated sections stay quiet.
Hidden / snoozed	User temporarily dismisses MILO	No hover activation until configured wake condition or explicit hotkey.
Fullscreen-suppressed	Fullscreen app/game or presentation detected where feasible	MILO stays hidden unless user explicitly invokes it.

6.4 Interaction priority

Explicit click, keyboard activation, and drag intent outrank hover. Hover must never cause a destructive action. If a user is typing in another app, MILO must not steal keyboard focus merely because the pointer passes over the island.

7. Information architecture

7.1 Top-level structure

Area	Purpose	Core content
Island shell	Entry and status cue	MILO mark/shape, subtle status indicators, expand affordance
Overview	Immediate glance	Time/date, battery/power state, now-playing summary, quick actions
System deck	At-a-glance system state and controls	Volume, brightness where supported, connectivity shortcuts/status, Bluetooth status, energy saver status
Productivity deck	Lightweight personal organisation	To-do lists, reminders, timers, alarms
Dropzone deck	Temporary file staging	Held files, reorder/remove/clear, drag-out affordance
Music deck	Media context and control	Artwork, title, artist, playback controls, progress, lyrics when available
Companion layer	Personality and feedback	Avatar expression/idle state and contextual reactions
Settings	Configuration and management	Appearance, behaviour, shortcuts, startup, privacy, data, diagnostics, updates

7.2 Navigation model

- Primary sections are reachable through a compact tab, rail, or clearly labelled navigation strip; final layout is a design decision validated by prototype testing.
- Overview is the default expanded landing surface.
- Section navigation must preserve relevant state, such as a running timer or held files.
- Vertical scrolling is supported for long sections; scroll boundaries must be visually and behaviourally clear.
- Do not hide essential controls behind hover-only secondary menus.

7.3 Information hierarchy

Priority order: immediate status and high-frequency action; current task context; secondary controls; configuration and less-frequent utilities. The deck must not become a dense grid of equally weighted widgets.

8. Global interaction and shell requirements

8.1 Placement and geometry

- UI-SHELL-001: The island MUST anchor to the top-center of the primary display by default.
- UI-SHELL-002: The island MUST remain within the usable work area and must not be clipped by display edges, notches, unusual scaling, or taskbars.
- UI-SHELL-003: Expanded content MUST grow downward from the top edge; any upward extension is reserved for a compact top-edge visual transition and must not cover the island's own anchor.
- UI-SHELL-004: The expanded deck MUST use responsive width and height limits, with internal scrolling when content exceeds available space.
- UI-SHELL-005: The user MUST be able to reposition the island or choose a supported anchor position if the design provides that setting; default remains top-center.
- UI-SHELL-006: On display topology changes, MILO MUST recalculate placement and remain reachable.

8.2 Expand and collapse

- UI-SHELL-010: Hovering over the collapsed island SHOULD expand it after a short configurable dwell threshold.
- UI-SHELL-011: The hover threshold MUST avoid accidental activation during pointer transit; a deliberate click or hotkey MUST open immediately.
- UI-SHELL-012: The deck MUST collapse after pointer exit only when the user is not interacting with it and no pin/lock state is active.
- UI-SHELL-013: Pointer movement from island to deck MUST not trigger collapse during the transition.
- UI-SHELL-014: Escape MUST dismiss an unpinned expanded deck; it MUST NOT cancel a running timer or discard data.
- UI-SHELL-015: A visible pin/keep-open control MUST be available when expanded.
- UI-SHELL-016: Hover alone MUST NOT steal keyboard focus from the foreground application.

8.3 Focus, stacking, and window behaviour

- UI-SHELL-020: The deck MUST appear above ordinary desktop windows when intentionally invoked, subject to Windows foreground and topmost rules.
- UI-SHELL-021: MILO MUST NOT permanently block clicks or keyboard interaction with underlying applications when collapsed.
- UI-SHELL-022: When expanded by pointer hover, MILO SHOULD support non-activating presentation where possible; controls become interactive on deliberate pointer/click action.
- UI-SHELL-023: Clicking outside an unpinned deck SHOULD dismiss it, except when the click is part of a drag/drop operation involving MILO.
- UI-SHELL-024: Focus order MUST be logical and visible; returning focus to the prior application MUST be predictable.
- UI-SHELL-025: The user MUST have a keyboard-accessible open/close mechanism.

8.4 Keyboard and pointer controls

Action	Default behaviour	Configuration
Hover island	Expand after dwell	Dwell duration and hover expansion on/off
Click island	Toggle expanded state	Hotkey alternative
Escape	Close unpinned deck	Fixed safety behaviour
Global shortcut	Open/close MILO	User configurable; conflict detection required
Tab / Shift+Tab	Move through controls	Fixed logical focus order
Enter / Space	Activate focused control	Standard control behaviour
Arrow keys	Operate sliders and list navigation where appropriate	Standard control behaviour
Drag file onto dropzone	Add file to temporary holding area	User gesture required
Drag held item outward	Begin native drag-out	User gesture required

8.5 Touch and precision input

MILO is designed primarily for mouse, keyboard, and touchpad, but touch targets SHOULD be comfortably sized on touch-enabled Windows devices. Hover-only functionality must have an equivalent click, keyboard, or touch activation path.

9. System deck requirements

9.1 Overview and time

- SYS-001: Show the current local time using the Windows system clock.
- SYS-002: Show date in a user-selectable or locale-appropriate format.
- SYS-003: Respect Windows regional settings, 12/24-hour preference, and timezone changes.
- SYS-004: Update time accurately without continuous high-frequency polling.
- SYS-005: Optional secondary time zones are out of initial release scope unless later prioritised.

9.2 Battery and power

- SYS-010: Display battery percentage when the device exposes a battery.
- SYS-011: Display charging, plugged-in, discharging, or unavailable state where available.
- SYS-012: Display battery saver / energy saver status when available.
- SYS-013: For desktops or systems without a battery, replace the percentage with a neutral power-state treatment or omit the battery widget without leaving an empty placeholder.
- SYS-014: Battery status must be labelled; colour alone cannot communicate low battery.
- SYS-015: MILO MUST NOT silently change Windows power plans or energy settings.

9.3 Volume

- SYS-020: Provide a master output volume control only where a reliable supported Windows integration exists.
- SYS-021: Show the current value and mute state when available.
- SYS-022: Slider changes MUST provide immediate feedback and avoid excessive native API calls through debouncing.
- SYS-023: If output device selection is not supported in the initial release, clearly open the relevant Windows sound settings rather than presenting a non-functional selector.
- SYS-024: MILO must not unexpectedly unmute audio.

9.4 Brightness

- SYS-030: Provide display brightness control only for supported internal displays/devices.
- SYS-031: Detect unsupported external monitors or devices and show a clear fallback to Windows display settings.
- SYS-032: Brightness changes MUST be reflected in the UI and confirmed by the native control layer.
- SYS-033: Do not simulate a hardware brightness change with a visual overlay and label it as actual system brightness.
- SYS-034: Multi-monitor brightness selection is a later enhancement unless validated across target hardware.

9.5 Connectivity and Bluetooth

- SYS-040: Show Wi-Fi/network connectivity status where the Windows integration exposes it reliably.
- SYS-041: Show Bluetooth state when available.
- SYS-042: If MILO cannot safely or consistently toggle a connectivity setting on the target Windows version, provide a shortcut to the relevant Windows Quick Settings page rather than faking a toggle.
- SYS-043: Never disconnect, pair, forget, or modify a device without explicit user action.
- SYS-044: Do not display saved network names or device identifiers unless required and explicitly enabled.

9.6 Control consistency

Every system control must distinguish between a status indicator, a direct control, and a shortcut that opens Windows settings. Unsupported states must be visible and honest. Controls should include tooltips or accessible names and clear pressed/selected states.

10. Productivity deck requirements

10.1 To-do lists

- PROD-001: Users can create, rename, and delete lists.
- PROD-002: Users can add, edit, complete, uncomplete, reorder, and delete tasks.
- PROD-003: Tasks support a title and optional note; due date/time and priority are optional fields.
- PROD-004: Completed tasks can be hidden, shown, or cleared with confirmation where deletion is irreversible.
- PROD-005: The overview may show a concise count or a small subset of upcoming/incomplete tasks.
- PROD-006: Task entry must be optimised for quick capture: one clear input and a low-friction save action.
- PROD-007: Data is stored locally and survives application restarts.

10.2 Reminders

- PROD-010: Users can create one-time reminders with title, optional note, and scheduled date/time.
- PROD-011: Reminders support edit, dismiss, snooze, and delete actions.
- PROD-012: A reminder notification must identify the reminder and offer an appropriate action to open or dismiss it.
- PROD-013: Past-due reminders must be handled predictably after sleep, shutdown, or app restart.
- PROD-014: The product must clearly explain that reminder delivery depends on Windows notification settings and device availability.

10.3 Timers

- PROD-020: Users can start a timer through preset durations and a custom duration.
- PROD-021: Timer states include idle, running, paused, completed, and dismissed.
- PROD-022: Users can pause, resume, reset, and cancel a timer.
- PROD-023: A running timer remains visible in a compact indicator when the deck is collapsed.

- PROD-024: Timer completion produces an in-app cue and, where permitted, a Windows notification/sound.
- PROD-025: Timer accuracy must be based on a monotonic or OS-supported elapsed-time source and must account for suspend/resume.

10.4 Alarms

- PROD-030: Users can create one-time and repeating alarms.
- PROD-031: Repeating alarms support day-of-week selection.
- PROD-032: Users can enable, disable, edit, dismiss, and delete alarms.
- PROD-033: Alarm delivery must communicate whether it is guaranteed while MILO is closed, the device is asleep, or the app is not running; do not promise delivery beyond validated platform capability.
- PROD-034: Snooze duration is configurable or selected from a small set of clear options.

10.5 Productivity rules

- PROD-040: No task, reminder, timer, or alarm may be silently discarded during navigation or collapse.
- PROD-041: Destructive actions require confirmation or a recoverable undo path.
- PROD-042: Empty states must explain how to add the first item.
- PROD-043: All productivity actions must be keyboard accessible.
- PROD-044: A user can use productivity tools without signing in or creating an online account.

11. Dropzone / temporary file holding requirements

11.1 Purpose

The dropzone is a temporary staging area: users can drop one or more files into MILO, keep them visible while switching applications, and drag them out into another app or file destination. It is not a file storage product and must not silently upload, copy, move, or delete files.

11.2 Core requirements

- FILE-001: Accept files dragged from Windows File Explorer and supported desktop applications.
- FILE-002: Display each held item with a useful filename, file type cue, and optional size where available.
- FILE-003: Support multiple held items and preserve their order.
- FILE-004: Allow an item to be removed individually.
- FILE-005: Provide a clear-all action with confirmation or undo.
- FILE-006: Support drag-out to compatible applications using native Windows drag/drop semantics.
- FILE-007: Preserve the original file; holding a file must not move or delete it.
- FILE-008: Persist enough metadata to restore the staging list after a restart, while validating that each source path still exists.
- FILE-009: If a source file is moved, renamed, deleted, or becomes inaccessible, show a missing/unavailable state and allow removal.
- FILE-010: Communicate when a destination app does not accept the dragged item or when the drag operation cannot be completed.

11.3 User journey

Step	Expected behaviour
1. Capture	User drags file(s) onto the visible dropzone or an explicit add action opens a native file picker.
2. Confirm	MILO shows the staged file(s), count, and concise metadata.
3. Switch context	The list remains available while the user moves to another app.
4. Retrieve	User drags one or more held items out of the dropzone.
5. Destination result	Destination app decides whether to copy, attach, import, or reject. MILO does not claim success until the native drag is accepted or the user receives clear feedback.
6. Cleanup	User removes items individually or clears the staging area.

11.4 Safety and edge cases

- Do not execute files, preview potentially unsafe content, or scan contents beyond what is required for display.
- Do not dereference arbitrary links or network paths without user action.
- Treat file names and metadata as untrusted display data; escape UI rendering.
- Handle duplicate files explicitly: allow duplicates or visually group them, but never silently overwrite an item.
- Support paths containing Unicode, spaces, long names, and network locations where Windows permissions allow.
- Do not promise drag-out compatibility with every app; document the tested application matrix.

11.5 Product decision

MILO's dropzone is a temporary reference to user-selected files, not a file copy operation. The source file remains where it is unless the destination application independently performs a copy or move.

12. Music and lyrics requirements

12.1 Now playing

- MEDIA-001: Detect the current system media session when available.
- MEDIA-002: Display title, artist, album, and artwork when provided by the media session.
- MEDIA-003: Provide play/pause, previous, next, and seek only when the active session exposes the relevant control.
- MEDIA-004: Reflect playback state and position without excessive polling.
- MEDIA-005: Support session changes and the case where multiple media sessions are present.
- MEDIA-006: If no session is available, show a helpful empty state rather than stale metadata.

12.2 Lyrics

- MEDIA-010: Display lyrics only when a permitted and supported source provides them.
- MEDIA-011: Clearly distinguish synced lyrics from unsynced text.
- MEDIA-012: If timestamped lyrics are available, highlight the current line and keep the current line visible.
- MEDIA-013: If lyrics are unavailable, show a neutral unavailable state; do not scrape arbitrary websites or bypass access controls.
- MEDIA-014: Attribute the lyrics source where required by its terms.
- MEDIA-015: Do not store, redistribute, or expose full lyric text beyond the applicable provider licence and caching rules.

12.3 Provider and rights constraints

Lyrics availability is a dependency, not a guaranteed platform primitive. Product and engineering must select a provider only after reviewing API access, licensing, attribution, geographic availability, rate limits, privacy terms, and commercial use rights. The feature must be modular so the music deck remains useful without lyrics.

12.4 Media failure states

- Player unavailable or session not exposing metadata.
- Player exposes metadata but not transport controls.
- Artwork unavailable or invalid.
- Track changes while lyrics are loading.
- Network unavailable for lyric retrieval.
- Provider quota/rate limit or authentication failure.
- User disables lyrics or network access.

13. MILO companion / avatar requirements

13.1 Role

The avatar gives MILO a recognisable personality and emotional warmth. It is not a chatbot, surveillance agent, or required part of task completion. It appears only when the deck is expanded, consistent with the agreed product direction.

13.2 Visibility and behaviour

- AV-001: The avatar MUST NOT appear in the collapsed island state.
- AV-002: The avatar MUST appear in the expanded deck in a dedicated, non-obstructive region.
- AV-003: The avatar MUST not cover controls, text, file items, or notifications.
- AV-004: Idle animation must be subtle and low-frequency.
- AV-005: Reactions may occur for meaningful events such as completing a timer or adding a task, but must be brief and non-blocking.
- AV-006: The avatar must not demand attention, speak automatically, or produce unsolicited messages.
- AV-007: Users must be able to reduce or disable avatar animation; a later preference may allow hiding the avatar entirely.

13.3 Expression system

State / event	Allowed response	Constraints
Idle	Breathing, blink, small posture shift	Very low motion; no rapid loops
Deck opened	Brief wake-up / welcoming motion	One-shot, short, no delay to interaction
Task completed	Small positive reaction	No interruption or blocking overlay
Timer completed	Attentive / celebratory cue	Must not replace the actual timer alert
Error	Neutral concerned expression	Never shame or blame the user
No network / unsupported feature	Neutral resting state	No misleading emotional response

13.4 Tone

The avatar's personality should feel friendly, calm, slightly playful, and respectful. Avoid infantilising language, guilt-based nudges, streak pressure, or emotional manipulation.

14. Settings, onboarding, and preferences

14.1 First-run onboarding

- SET-001: First launch introduces MILO's island, how to expand/collapse it, and how to pin it open.
- SET-002: Explain the primary sections in a short, skippable sequence.
- SET-003: Explain startup behaviour and notification requirements without preselecting intrusive options.
- SET-004: Request permissions only at the moment they are needed and explain why.
- SET-005: Provide a skip path and a way to revisit tips from Settings.
- SET-006: Do not require account creation.

14.2 Settings categories

Category	Settings
General	Launch at startup, open behaviour, close/minimise behaviour, default display, app language if supported
Appearance	Theme/accent, island size, deck width, transparency/glow intensity, animation intensity
Interaction	Hover dwell, click-to-expand, collapse delay, pin behaviour, global shortcut
Productivity	Default timer presets, alarm snooze, notification preferences, completed-task visibility
Music	Media session preference, lyrics on/off, lyrics provider attribution/preferences where relevant
Dropzone	Persistence, maximum item count if imposed, clear-on-exit preference
Privacy	Local data overview, diagnostics opt-in, clear data, reset app
Accessibility	Reduced motion, larger text, contrast mode, keyboard hints
About & updates	Version, update channel, check for updates, licences, support, release notes

14.3 Startup and tray behaviour

- MILO SHOULD offer launch-at-login as an explicit preference.
- MILO SHOULD provide a system tray icon or equivalent background access point.
- Closing the visible deck/window must be clearly distinguished from exiting the application.
- Users must be able to fully quit MILO from its menu and prevent background execution.
- Startup failure must not create an infinite relaunch loop.

14.4 Preference persistence

Preferences must persist across restarts. Invalid or removed preference values must fall back to safe defaults. Import/export of settings is a possible later enhancement, not a launch requirement.

15. Notifications, reminders, and background behaviour

15.1 Notification principles

- Notifications are event-driven and limited to user-created reminders, timers, alarms, and essential application status.
- No marketing, promotional, or engagement notifications in the initial release.
- Respect Windows notification permission and focus/quiet modes to the extent exposed by the platform.
- Provide in-app history or clear state for missed notifications where useful.

- Notification text must avoid exposing sensitive notes by default on the lock screen; user preference may control detail.

15.2 Background operation

MILO may run a lightweight background process to maintain timers, schedule reminders, and monitor required system state. Background work must be bounded, observable in diagnostics, and reduced when the system is on battery saver or under resource pressure.

15.3 Sleep, shutdown, and restart

- Timers must reconcile elapsed time after sleep using stored target time and a reliable clock model.
- One-time reminders that become overdue while the app is not running must be shown on next launch according to a documented catch-up policy.
- Repeating alarms must calculate the next occurrence using local timezone rules and daylight-saving transitions.
- Do not imply that an alarm will wake a sleeping or powered-off PC unless that behaviour is explicitly implemented and validated.
- On unclean shutdown, local data must remain transactionally consistent.

16. Visual design system and motion

16.1 Visual identity

MILO uses a dark, luminous, rich and elegant futuristic style. It combines glassmorphism-inspired depth with a distinct material language: soft translucent layers, carefully controlled blur, luminous edges, organic silhouettes, subtle gradients, and precise typography.

16.2 Design tokens

Token family	Requirement
Base surfaces	Deep charcoal / ink backgrounds; avoid pure black as the only surface.
Elevated surfaces	Layered dark translucent panels with enough opacity for readable content.
Accent	A small, consistent accent palette with a primary luminous hue and restrained secondary gradient.
Text	High-contrast primary text, quieter secondary text, and clearly distinguishable disabled states.
Borders	Soft, low-contrast edge definition; stronger focus ring for keyboard navigation.
Radius	Soft organic curves; radii should be consistent across related components.
Shadows/glow	Use sparingly to define depth and focus, not as a permanent halo around every control.
Icons	Consistent line weight, optical alignment, and clear meaning; labels accompany ambiguous controls.
Typography	Legible modern sans-serif with hierarchy for time, section titles, metadata, and helper text.

16.3 Layout

- The collapsed island must remain visually simple and recognisable at a glance.
- The expanded deck should feel like one coherent object rather than a collection of unrelated cards.
- Use spacing and grouping to distinguish system, productivity, files, and music.
- Content density must be controlled; the overview should not expose every possible setting simultaneously.
- Responsive layout must preserve touch target size and text readability across common Windows scaling factors.

16.4 Motion

- Expansion and collapse should feel smooth, spatially connected, and quick.
- Motion must not delay control availability or create a perception of lag.
- Use short, purposeful transitions for hover, selection, tab change, and avatar reactions.
- Respect reduced-motion preference and Windows animation settings where feasible.
- Avoid parallax, constant pulsing, rapid bouncing, and long decorative loops.

16.5 Visual quality gates

- Text remains readable over all translucent surfaces and artwork.
- Controls have clear hover, pressed, focused, disabled, loading, and error states.
- Gradients never reduce text contrast or imply status without a label.
- Blur and transparency degrade gracefully when system composition or graphics capabilities are limited.
- All animation remains smooth on the minimum supported hardware target.

17. Accessibility and inclusive design

Accessibility is a product requirement, not a polish task. MILO must support users who do not use a mouse, cannot perceive colour distinctions, use screen readers, need larger text, or prefer reduced motion.

- A11Y-001: Every interactive control has a meaningful accessible name, role, and state.
- A11Y-002: All core features are operable using keyboard only.
- A11Y-003: Focus order follows the visual and task hierarchy; focus is always visible.
- A11Y-004: Normal text targets at least 4.5:1 contrast; large text targets at least 3:1.
- A11Y-005: Status is never communicated by colour alone.
- A11Y-006: The application remains usable with Windows high-contrast themes or provides an equivalent high-contrast mode.
- A11Y-007: Text scaling and display scaling do not clip controls or make content unreachable.
- A11Y-008: Reduced-motion mode suppresses nonessential transitions and avatar animation.
- A11Y-009: Screen reader announcements are concise and event-based; avoid announcing every clock tick or slider update.
- A11Y-010: Hover-triggered content has keyboard and click alternatives.

17.1 Accessibility test plan

- Run Windows accessibility inspection tools on all primary surfaces.
- Test with Narrator for reading order, control labels, notifications, and focus restoration.
- Test keyboard-only completion of each primary user journey.
- Test at common display scaling values and with larger text settings.
- Test high-contrast and reduced-motion settings.
- Include users with different input and perception needs in usability validation where feasible.

18. Privacy, security, and trust

18.1 Data principles

- Store personal productivity data locally by default.
- Do not require account creation or cloud synchronisation for core functionality.
- Do not collect file contents for the dropzone; retain only the metadata and source reference required to stage the item.

- Do not collect browsing history, keystrokes, screenshots, microphone, or camera data.
- Network requests must be limited to clearly disclosed features such as update checks or a lyrics provider.
- Diagnostics/telemetry, if introduced, must be opt-in, minimised, and documented.

18.2 Permissions and native access

Native integrations must use least privilege. The webview/frontend must receive only the specific command permissions needed for each feature. File access must be user-initiated and scoped. External URLs and update endpoints must be allowlisted and validated. The application must not execute arbitrary commands supplied by remote content.

18.3 Security requirements

- SEC-001: Validate and sanitise all data crossing the frontend/native boundary.
- SEC-002: Treat file names, media metadata, lyric text, and provider responses as untrusted input.
- SEC-003: Keep secrets and provider credentials out of the frontend bundle and logs.
- SEC-004: Use secure transport for all network calls.
- SEC-005: Sign release binaries and publish checksums or equivalent integrity information.
- SEC-006: Restrict update manifests and package sources to trusted endpoints.
- SEC-007: Avoid broad filesystem permissions when a user-selected file reference is sufficient.
- SEC-008: Provide a clear way to clear local data and exit the application.

18.4 Trust UX

When a feature needs permission, MILO must explain the specific capability, why it is needed, what data is accessed, whether data leaves the device, and how to revoke or disable the feature. Avoid vague permission language such as 'improve your experience'.

19. Reliability, performance, and quality attributes

19.1 Performance targets

Metric	Target / expectation
Cold start	Target under 3 seconds on reference mid-range Windows hardware; establish a reproducible benchmark.
Island response	Pointer activation should feel immediate; target under 150 ms before transition begins.
Expand/collapse	Target 180–320 ms transition, adjustable for reduced motion.
Idle CPU	Near-zero sustained CPU when idle; target average below 1% on reference system, excluding OS scheduling noise.
Idle memory	Target under 150 MB working set for the baseline app; profile and revise with actual build.
Interaction latency	Common control response should begin within 100 ms when native OS call is available.
Data persistence	User-created items survive normal restart and recover from interrupted writes.
Crash recovery	Application can restart without corrupting user data or repeatedly crashing on invalid persisted state.

19.2 Reliability

- Core shell and local productivity tools must function without an internet connection.
- System integration failure in one module must not crash the whole application.
- Every native command must return a structured success/failure result.

- Background timers and reminder scheduling must be tested across suspend/resume and timezone changes.
- Logs must be useful for debugging while avoiding sensitive content.

19.3 Compatibility

The final supported Windows version range must be explicitly chosen before beta. Technical feasibility should be validated on Windows 10 and Windows 11 variants, with the media-session API baseline and any packaging/runtime dependencies documented. Hardware-dependent controls must be tested on laptops, desktops, external displays, and devices with/without Bluetooth or battery.

19.4 Multi-monitor and DPI

- Handle monitor attach/detach, resolution change, scaling change, primary-display change, and display sleep/wake.
- Use logical and physical coordinate conversion correctly.
- Keep the island on the selected display and prevent off-screen placement.
- Ensure expanded deck geometry respects the active display work area.

20. Data lifecycle and user control

20.1 Data classes

Data	Storage expectation	User control
Preferences	Local configuration	Edit, reset, export/import later
Tasks and lists	Local database	Edit, delete, clear all
Reminders and alarms	Local database plus schedule metadata	Edit, dismiss, delete
Timer state	Local transient state plus persistence for restart reconciliation	Pause, reset, cancel
Dropzone entries	Local references and display metadata	Remove, clear; validate source on restore
Media metadata	Ephemeral in-memory state by default	Not retained beyond need
Lyrics	Ephemeral/cache only if provider terms allow	Disable lyrics; clear cache where applicable
Diagnostics	Local logs with rotation	View/export/delete

20.2 Deletion and reset

- Users can delete individual tasks, reminders, alarms, and staged files.
- Clear-all actions must state exactly what will be removed.
- Reset application must explain whether it removes preferences, productivity data, cached data, and logs.
- Uninstall must not leave personal data behind without a clear disclosure; offer an explicit retain-data option only if supported.

20.3 Backup and sync

Cloud sync and automatic backup are out of initial scope. The local data model should be designed so that future export/import or sync can be added without changing the user-facing semantics of core records.

21. Error handling and edge cases

21.1 Error principles

- Explain what happened in plain language.
- Say what remains safe and what the user can do next.
- Do not expose stack traces or raw native errors in normal UI.
- Do not claim a change succeeded until the native layer confirms it.

• Preserve user data and allow retry where safe.

21.2 Required edge-case coverage

Area	Cases to handle
Shell	Pointer exits during animation; deck overlaps taskbar; fullscreen app; focus changes; display removed; scaling changed; hotkey conflict
System	No battery; no supported brightness control; no Bluetooth adapter; network unavailable; permission denied; device state changes mid-action
Productivity	Duplicate titles; invalid date/time; daylight-saving transition; reminder missed while closed; app suspended; corrupt record
Dropzone	File deleted/moved; permission denied; network path unavailable; duplicate item; long Unicode filename; destination rejects drop; multi-file drag
Music	No session; multiple sessions; metadata absent; control unsupported; track changes mid-request; provider offline; lyrics unavailable
Avatar	Animation disabled; asset fails to load; reduced motion; low-power mode; user hides avatar
Updates	Offline; invalid manifest; signature mismatch; insufficient disk; update interrupted; rollback needed

21.3 Empty and loading states

Every module must define loading, empty, unavailable, permission-denied, and error states. Skeleton loading should be brief and purposeful; avoid endless spinners. Empty states should give one clear next action.

22. Analytics and success measurement

22.1 Measurement philosophy

MILO should be measurable without becoming invasive. Initial product validation can rely on usability studies, opt-in feedback, crash diagnostics, and privacy-preserving aggregate events. Do not collect file names, task text, reminder content, lyrics, or media listening history.

22.2 North-star metric

Weekly successful utility actions per active user, measured only with consent and without collecting the content of those actions.

22.3 Supporting metrics

Metric	Definition / interpretation
Activation	Percentage of new installs that complete first-run setup and successfully expand MILO
Time to first value	Time from first launch to first successful meaningful action
Feature adoption	Share of active users who use system, productivity, dropzone, or music modules
Repeat use	Weekly active usage and returning-user rate
Task success	Completion rate for core journeys in usability tests
Interaction friction	Mis-triggered expansion, accidental dismissals, failed drags, and repeated retries
Reliability	Crash-free sessions, recovery success, data-loss incidents
Performance	Startup, expand latency, idle CPU/memory, and control response
Trust	Permission opt-out, uninstall feedback, privacy concerns, and support issue trends

22.4 Qualitative research

- Observe first-run discoverability without coaching.
- Ask whether the island is helpful or distracting during real work.
- Test whether the dropzone mental model is understood and whether users trust that files remain in place.
- Evaluate whether avatar reactions feel charming, neutral, or intrusive.
- Identify which module earns repeat use and which should be simplified.

23. Scope, prioritisation, and release strategy

23.1 Priority definitions

Priority	Meaning
P0 — Must ship	Core product identity or safety/reliability requirement; release blocked if missing.
P1 — Should ship	Important expected capability; may be deferred only with explicit product approval.
P2 — Later	Useful enhancement that is not required to prove the core product.
Out of scope	Explicitly excluded from initial release.

23.2 Initial release scope

Module	Priority	Release expectation
Island shell and expansion	P0	Polished, reliable, multi-monitor aware, keyboard accessible
Overview/time	P0	Accurate local time/date and useful empty/loading states
Battery/power status	P0	Battery and saver status where supported; graceful no-battery state
Volume control	P1	Ship if reliable native control integration passes device matrix
Brightness control	P1	Ship only on validated supported devices; otherwise settings fallback
Wi-Fi/Bluetooth status/shortcuts	P1	Honest status and safe OS settings fallback
To-do lists	P0	Create/edit/complete/delete, local persistence
Timers	P0	Reliable lifecycle, notifications, suspend/resume reconciliation
Reminders	P0	Create/schedule/dismiss/snooze with documented delivery limits
Alarms	P1	One-time/repeating with validated background semantics
Dropzone	P0	Stage, persist references, drag-out, remove/clear
Now playing	P1	Current session metadata and supported transport controls
Lyrics	P2	Provider and rights validated; optional module
Avatar	P1	Expanded-only, subtle, optional/reduced motion
Settings/onboarding	P0	Core preferences, privacy controls, clear onboarding
Installer, signing, updater	P0	Trusted install, update, uninstall and recovery

23.3 Suggested milestones

Milestone	Exit outcome
M0 — Discovery & validation	User problem, visual prototype, Windows capability spike, risk register
M1 — Shell foundation	Island positioning, expand/collapse, focus, DPI, tray, settings persistence
M2 — Core utility	Overview, time, power, tasks, timers, reminders
M3 — System & file workflows	Validated system controls, dropzone capture and drag-out
M4 — Media & personality	Media session, avatar, lyrics decision/provider integration if approved
M5 — Hardening	Accessibility, security review, performance, edge cases, installer/update
M6 — Beta & release	Usability validation, crash-free criteria, release notes, support readiness

23.4 Scope change control

Any proposed feature that expands system permissions, adds a cloud dependency, introduces user-content collection, or materially changes the island interaction model requires explicit product and security review. Changes to P0 requirements require an impact statement covering usability, engineering effort, QA, privacy, and release date.

24. Acceptance criteria and QA strategy

24.1 Core acceptance criteria

- AC-001: On launch, MILO appears at the configured top-edge anchor and remains within the selected display bounds.
- AC-002: Hover and explicit activation expand the deck without stealing focus from unrelated foreground typing.
- AC-003: The deck collapses predictably and never discards unsaved user input.
- AC-004: Time/date reflect Windows locale and timezone settings.
- AC-005: Battery/power indicators correctly handle battery and no-battery devices.
- AC-006: Tasks and lists persist across app restart and support all core CRUD actions.
- AC-007: Timers remain accurate across pause/resume and suspend/resume test cases.
- AC-008: Reminders recover overdue state after restart according to the documented policy.
- AC-009: Dropzone can accept supported files and initiate drag-out without altering source files.
- AC-010: Media controls are enabled only when the active session exposes those controls.
- AC-011: Avatar is absent when collapsed and never blocks core interaction when expanded.
- AC-012: Keyboard-only user can access and operate every P0 feature.
- AC-013: Clear-data and quit actions behave as documented.
- AC-014: Installer, update, and uninstall flows pass on clean and upgrade scenarios.

24.2 Test layers

Layer	Coverage
Unit	Time calculations, recurrence, state transitions, validation, persistence models, permission decisions
Integration	Native command contracts, Windows API wrappers, DB transactions, notifications, media session
UI	Layout, responsive behaviour, hover/click/keyboard, focus, animations, empty/error states
System	Sleep/wake, display topology, DPI, battery saver, Windows notification settings, startup
Compatibility	Supported Windows builds, hardware matrix, external displays, touchpad/mouse/keyboard
Accessibility	Narrator, keyboard-only, contrast, text scaling, reduced motion
Security	Input validation, file path handling, update integrity, permissions, network boundaries
Performance	Startup, idle CPU/memory, transition latency, long-running soak tests

24.3 Release gates

- No unresolved critical or high-severity security issue.
- No known data-loss defect in tasks, reminders, timers, or staged-file metadata.
- All P0 acceptance criteria pass on the supported Windows matrix.
- Accessibility baseline passes keyboard, screen reader, contrast, and reduced-motion checks.
- Crash-free beta threshold is met and startup/idle performance is within agreed budgets.
- Installer, signing, update, rollback/recovery, and uninstall are verified.
- Support documentation explains limitations for hardware controls, alarms, and lyrics.

25. Dependencies, risks, and mitigations

Risk / dependency	Impact	Mitigation
Windows API variability	Some controls differ by OS version or hardware	Capability detection, device matrix, honest fallback, modular native adapters
Brightness support inconsistency	Slider may not work on external displays	Use documented native interfaces where supported; never fake hardware state
Media provider inconsistency	Metadata and transport support vary	Capability-aware controls, session selection, empty states
Lyrics licensing/provider access	Lyrics may be unavailable or commercially restricted	Treat as optional; legal/provider review before launch
Background alarm guarantees	Windows sleep/shutdown can prevent delivery	Precise product language, catch-up policy, platform validation
Drag/drop interoperability	Some destination apps reject shell file drops	Test representative apps, preserve source, provide fallback open/reveal action
Overlay focus conflicts	Could disrupt typing or fullscreen apps	Non-activating hover, explicit activation, suppression, focus regression tests
GPU/transparency cost	Blur and animation may affect performance	Reduced effects mode, profiling, fallback surfaces
Privacy concerns	Users may distrust a persistent desktop utility	Local-first defaults, transparent permissions, no covert collection
Feature overload	Deck becomes cluttered and difficult to navigate	Progressive disclosure, prioritise overview, usability testing, P0/P1 discipline

25.1 External dependencies

- Windows native APIs and runtime support for system status, media sessions, notifications, and window behaviour.
- Appropriate packaging and code-signing infrastructure.
- Optional lyrics provider, subject to rights, terms, access, and service continuity.
- Design assets and animation files for the companion.
- Automated build, test, signing, release, and update pipeline.

26. Open decisions and validation plan

26.1 Decisions to lock before implementation freeze

Decision	Why it matters	Validation required
Minimum supported Windows version	Determines API availability and packaging	Technical spike on Windows 10/11 builds; publish explicit matrix
Exact island dimensions and hover dwell	Core product identity and accidental activation risk	Interactive prototype and user testing
Primary navigation pattern	Determines deck density and discoverability	Compare tabs, rail, and scrollable section model
Global hotkey default	Potential conflict with other apps	Conflict detection and user-configurable default
System control implementation scope	Hardware support varies	Device compatibility matrix before promising controls
Alarm behaviour when app is closed/asleep	User trust and safety	Validate OS scheduling/notification architecture and document guarantee
Lyrics provider	Legal, commercial, technical dependency	Provider terms, licence, geography, API and cost review
Avatar visual design and animation intensity	Brand character and distraction	Prototype testing with animation on/off
Dropzone persistence policy	Convenience vs stale file references	Test restore semantics and user expectations

26.2 Validation plan

- Conduct concept tests on the island metaphor and discoverability.
- Prototype expansion, collapse, pinning, and pointer travel to detect accidental dismissals.
- Test core workflows with at least a small mixed group of focused workers, students, and power users.
- Run native integration spikes before finalising promises for brightness, volume, Bluetooth, and alarms.
- Test file drag-out against representative browser upload controls, chat apps, design tools, and office apps.
- Validate music session handling across common Windows media players and browsers.
- Review lyrics licensing before building a dependency into the release plan.

26.3 Decision ownership

Product owner approves user-facing scope and priorities. Engineering lead approves feasibility and architecture. Design lead approves interaction and visual consistency. QA lead approves test coverage and release evidence. Security/privacy reviewer approves permission and data-handling changes.

Appendix A — Requirement ID index

Prefix	Domain
UI-SHELL	Island shell, window behaviour, expansion, focus, input
SYS	System overview and native quick controls
PROD	Tasks, reminders, timers, alarms
FILE	Dropzone and staged file references
MEDIA	Media sessions, transport, lyrics
AV	Avatar and companion behaviour
SET	Settings and onboarding
A11Y	Accessibility
SEC	Security and privacy
AC	Acceptance criteria

Traceability rule

Each requirement ID must map to at least one design artefact, implementation issue, and test case before release. Any requirement removed or changed must be recorded with rationale and an approval owner.

Appendix B — Key user journeys

Journey 1: Quick glance and dismiss

- User moves pointer to island; dwell threshold is reached.
- MILO expands without taking focus away from the user's active application.
- User checks time, battery, or now-playing information.
- User moves pointer away; deck collapses after configured delay.

Journey 2: Capture a reminder

- User opens MILO using hover, click, or hotkey.
- User navigates to Productivity and selects Add reminder.
- User enters title and date/time; optional note is added.
- User saves; MILO confirms creation and returns to the relevant list.
- At due time, MILO presents the reminder through the configured notification path.

Journey 3: Stage and drag a file

- User expands MILO and drags a file from Explorer onto the dropzone.
- MILO displays the file as staged and keeps the original in place.
- User switches to a browser or other destination app.
- User drags the staged item into the destination's supported drop target.
- MILO retains the item until the user removes it or the configured cleanup policy applies.

Journey 4: Control current music

- Music is playing in a supported Windows media session.
- MILO shows track metadata and artwork in the overview/music deck.
- User pauses or skips using an available transport control.

- If lyrics are available from a permitted provider, MILO shows them with attribution and sync behaviour where supported.
- If not available, the music controls continue to work without lyrics.

Journey 5: Run a timer

- User chooses a preset or enters a custom duration.
- Timer starts and a compact running indicator remains available when MILO collapses.
- User can pause, resume, reset, or cancel.
- On completion, MILO provides the configured alert and a clear dismiss/snooze action.

Appendix C — Glossary

Term	Definition
Island	The compact top-edge entry point for MILO.
Deck	The expanded interactive surface revealed from the island.
Collapsed state	The compact resting state.
Expanded state	The visible interactive deck.
Pinned state	A user-selected state that keeps the deck open.
Dropzone	Temporary holding area for references to files selected by the user.
Media session	A Windows-exposed playback session from another application.
Synced lyrics	Lyrics with timestamps that can be highlighted in alignment with playback.
Local-first	Core functionality stores data on the user's device by default.
Graceful degradation	A feature provides a clear fallback when a capability is unsupported or unavailable.

Appendix D — External references

The following current platform references informed the product constraints in this PRD. They are implementation references, not a substitute for device-level validation.

1. Tauri v2 — Capabilities and permissions
https://tauri.app/security/capabilities/
Permission boundaries for frontend/webview access.

2. Tauri v2 — Global Shortcut plugin
https://v2.tauri.app/plugin/global-shortcut/
Global shortcut registration and explicit permission model.

3. Tauri v2 — Window customization
https://v2.tauri.app/learn/window-customization/
Custom window decoration and window permission considerations.

4. Tauri v2 — Configuration reference
https://v2.tauri.app/reference/config/
Window configuration and platform-specific properties.

5. Microsoft Learn — GlobalSystemMediaTransportControlsSessionManager
https://learn.microsoft.com/en-us/uwp/api/windows.media.control.globalsystemmediatransportcontrolssessionmanager
Windows media session discovery and current-session behaviour.

6. Microsoft Learn — Windows.Media.Control namespace
https://learn.microsoft.com/en-us/uwp/api/windows.media.control
Media playback session metadata and controls.

7. Microsoft Learn — EnergySaverStatus
https://learn.microsoft.com/en-us/uwp/api/windows.system.power.energysaverstatus
Battery saver state and resource-aware behaviour.

8. Microsoft Learn — Backlight Control Interface
https://learn.microsoft.com/en-us/windows/win32/power/backlight-control-interface

Documented display backlight control interface.

9. Microsoft Learn — Accessibility checklist for Windows apps
https://learn.microsoft.com/en-us/windows/apps/design/accessibility/accessibility-checklist
Accessible names, keyboard support, contrast, and testing.

10. Microsoft Learn — Accessible text requirements
https://learn.microsoft.com/en-us/windows/apps/design/accessibility/accessible-text-requirements
Contrast and semantic text roles.

Reference review date: 02 November 2026. Platform APIs and service terms may change; engineering must re-check the relevant documentation before implementation and release.

Appendix E — Product readiness checklist

- Product vision, target audience, goals, and non-goals approved.
- Minimum Windows version and supported hardware matrix published.
- All P0 requirements have design, implementation, and QA traceability.
- Prototype validates discoverability and non-intrusive expansion.
- System control support and fallback behaviour validated on target devices.
- Alarm and reminder delivery semantics documented honestly.
- Dropzone behaviour validated across representative source and destination applications.
- Media metadata and control support validated across representative media sessions.
- Lyrics provider and rights review completed or feature deferred.
- Accessibility checks passed, including keyboard, Narrator, contrast, scaling, and reduced motion.
- Privacy disclosures and permission prompts reviewed.
- Installer, signing, update, recovery, and uninstall paths tested.
- Support guidance and release notes prepared.